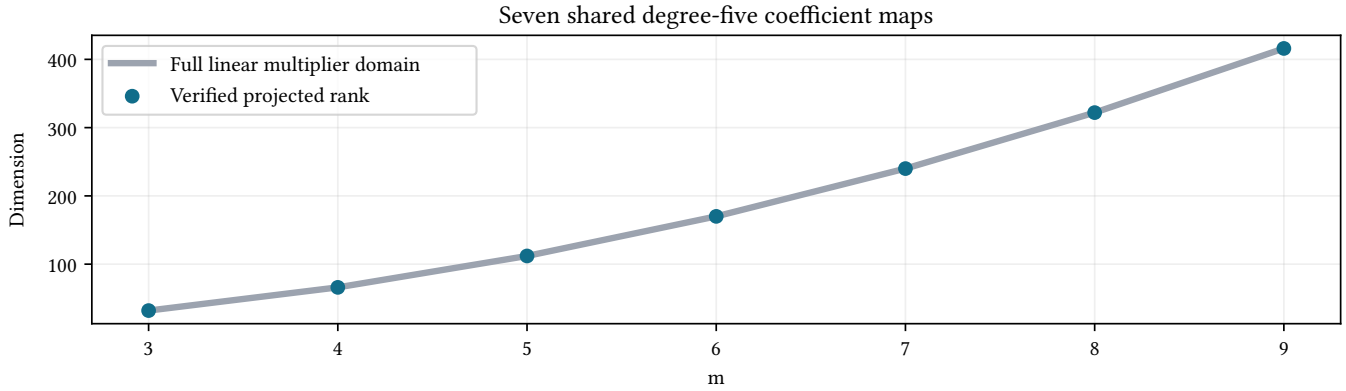
Prime degree-five syzygy exclusion

All42 retained prime m3–9 cases passed producer,fresh andS3-restored replay. Seven shared templates; independent scientific review pending.

m	Cases	Rank / columns	FFD lower
3	6	32 / 32	≥6
4	6	66 / 66	≥6
5	6	112 / 112	≥6
6	6	170 / 170	≥6
7	6	240 / 240	≥6
8	6	322 / 322	≥6
9	6	416 / 416	≥6



Meaning. Six overlapping five-selector restrictions cover all27 selectors in each block and retain all intermediate variables. Full projected rank excludes original linear syzygies. Selector squares form a regular sequence; the written quotient argument gives the lower bound. A deficient projection is inconclusive. Actual fixed factor bases and original equations are unchanged.

Combined evidence. Earlier upperFFD17 certificates are joined only after source and base-manifest hashes match for all42 cases. ExactFFD remains unknown. No newLFD,solving-degree,elite-base or performance claim.

Checks. 21 test invocations comprise five new tests and duplicated eight donor tests. Six independent dense rank comparisons and30 independently expanded quartics; annihilator and coverage controls. All149 raw payloads verify before/after restored replay.

Execution and retention. One process,2GiB,300s producer/300s replay,720s outer,no retries. Existing-host fallback after retained EKS source403. Rejected recovery-note headings and correction retained; gate passed before production. Interval:2026-10-30T07:12:06Z to2026-10-30T07:12:13Z,diagnostic only.

Replay. PRIME_LINEAR_FORMAT.json documents every field. Extract archive,check manifest,run `instrument replay ROOT ROOT/protocol-prime-linear.json ROOT/results`,check manifest again. All original and new evidence remains retained. Full three-family m2–9 and elite-base objective open.

Source:acbf026e945b486e461262b8093c598375993cf3
Protocol:fab78c9ce1792ae5aea977903df296df49f591d575725905de4010fa806f7198
Data:8882bd015df59c087067fffcbb54606be253cc3c Caminata–Gorla Definition1.3 and Section4study argument:PRIME_LINEAR_DESIGN.md.